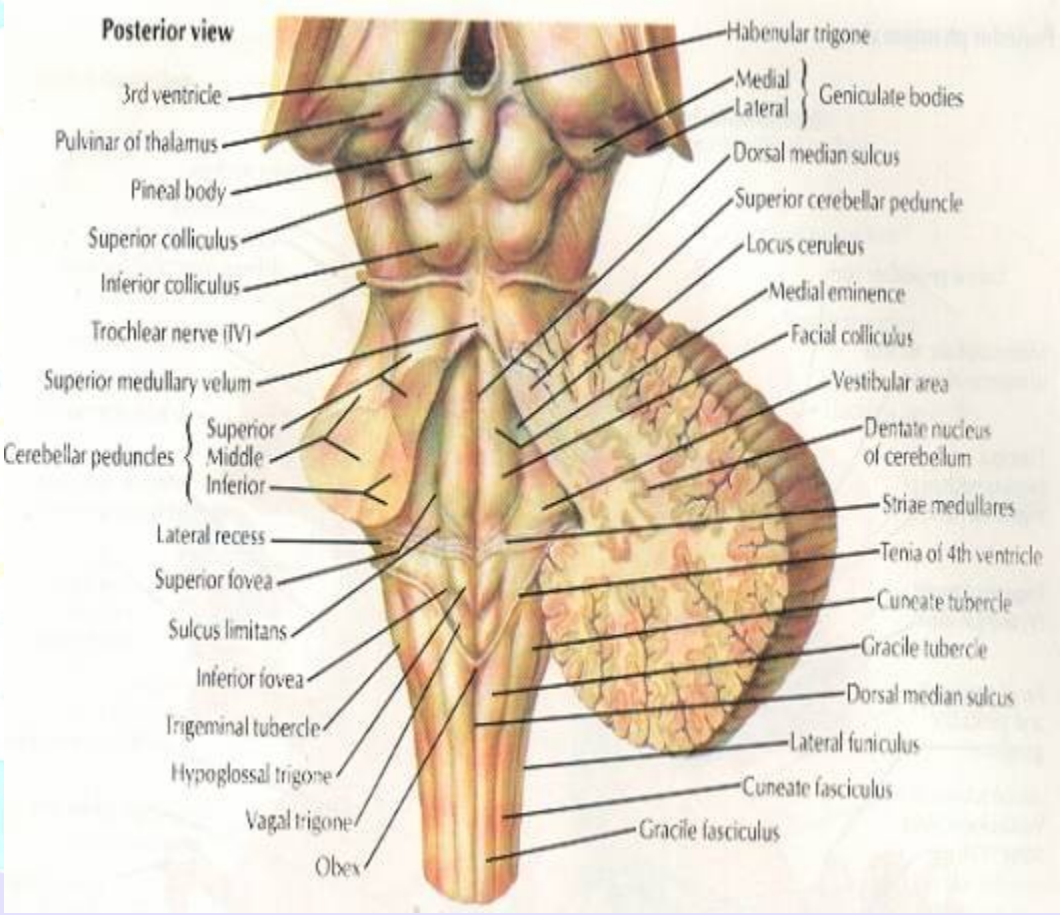
DIENCEPHALON II

EPITHALAMUS SUBTHALAMUS AND THIRD VENTRICLE

EPITHALAMUS

- Small region superior to the thalamus.
- Consists of pineal gland which secretes a hormone called melatonin.
- Melatonin induces sleep.
- **Anterior & posterior paraventricular nuclei**
- **Habenular nuclei**
- **Stria medullaris thalami**
- **Pineal body**

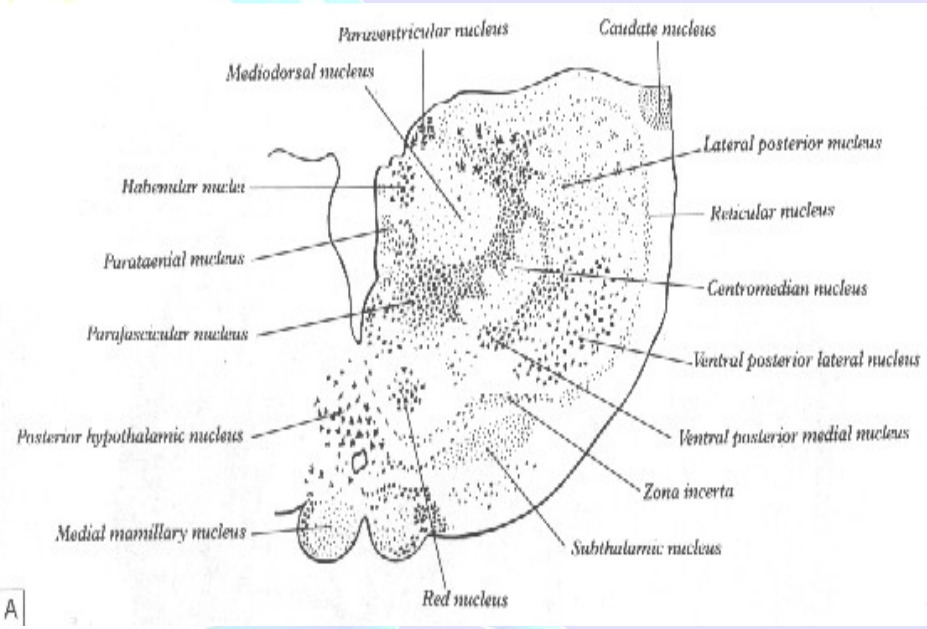


EPITHALAMUS CONNECTIONS:

- **PARAVENTRICULAR NUCLEI:**
- **Underlie ependyma of third ventricle**
- **Part of midline thalamic nuclei**

- **AFFERENTS:**
  - **From hypothalamus, septal nuclei**
  - **Stria terminalis, hippocampal formation**

- **EFFERENTS:**
  - **Nucleus Accumbens**
  - **Amygdyla**
  - **Hippocampal formation**

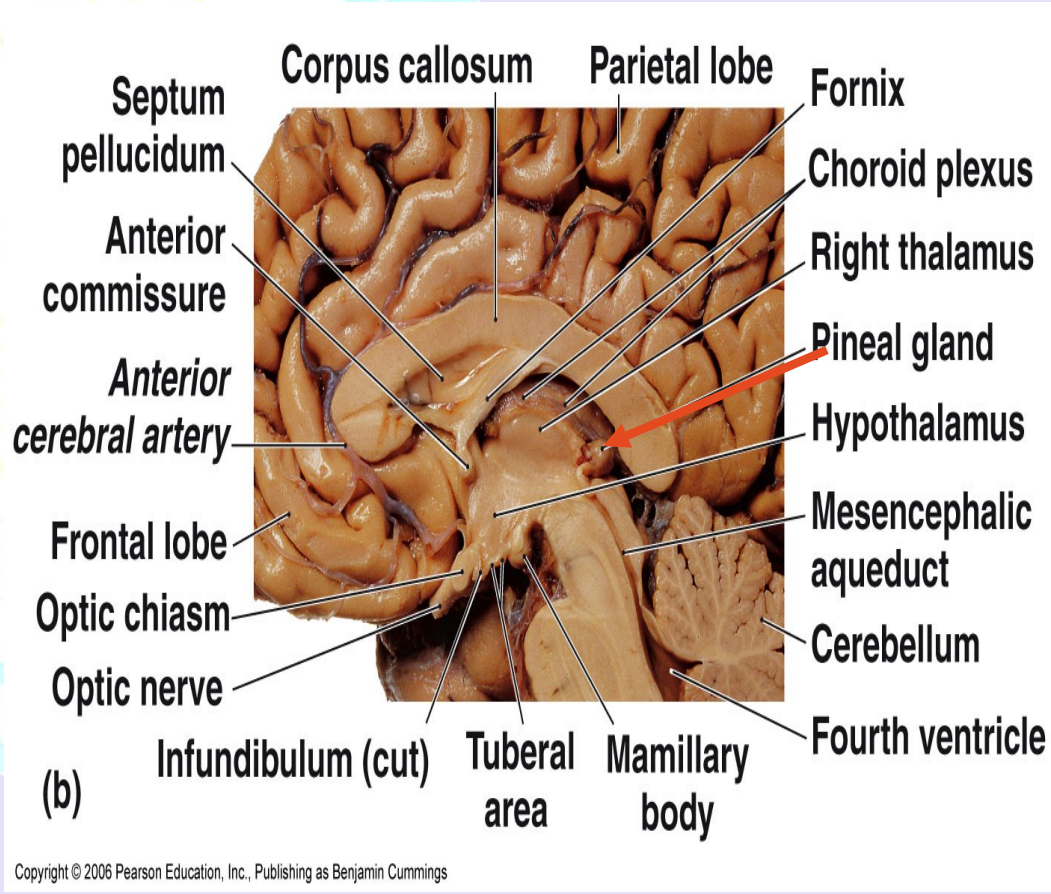


EPITHALAMUS CONNECTIONS:

- HABENULAR NUCLEI & STRIA MEDULLARIS
- Poteriorly, dorsomedial corner
- Two nuclei medial & lateral
- Habenular commisure
- AFFERENTS- in stria medullaris
  - Prepiriform cortex
  - Globus pallidus
  - Hypothalamus
  - midbrain raphe nucleus
  - lateral dorsal tegmental nucleus
  - Substantia nigra
- EFFERENTS
  - Interpeduncular nucleus- fasciculus retroflexus
  - Mediodorsal thalamic nucleus
  - Mesencephalic tectum
  - Midbrain raphe nucleus
  - reticular formation
- Ablation cause extensive changes in metabolism, endocrine & thermal regulation

EPITHALAMUS: PINEAL GLAND

- Indirect photosensitive neuroendocrine organ
- Cone shaped
- Attached to roof of third ventricle
- Pinealocytes
- Secretes: serotonin, norepinephrine & melatonin, also TRH, LHRH & somatostatin
- Melatonin synthesis are rhythmic & in direct response to the daily cycle of photic input
- Suprachiasmatic nuclei – circadian rhythms



EPITHALAMUS; PINEAL GLAND

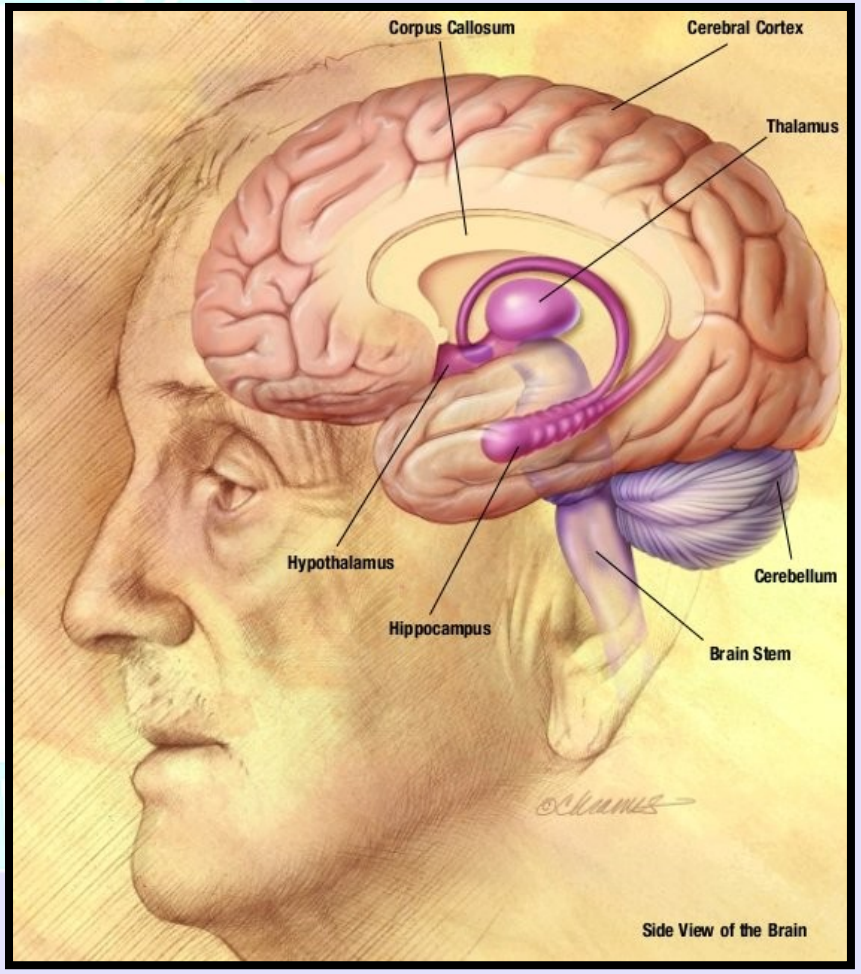
- **Parenchymatous pinealomas – depression of gonadal functin & delayed pubescence**
- **Lesions which destroy pineal gland – precocious puberty**
- **Inhibitory influence on gonads & reproductive system**

SUBTHALAMUS

SUB THALAMUS:

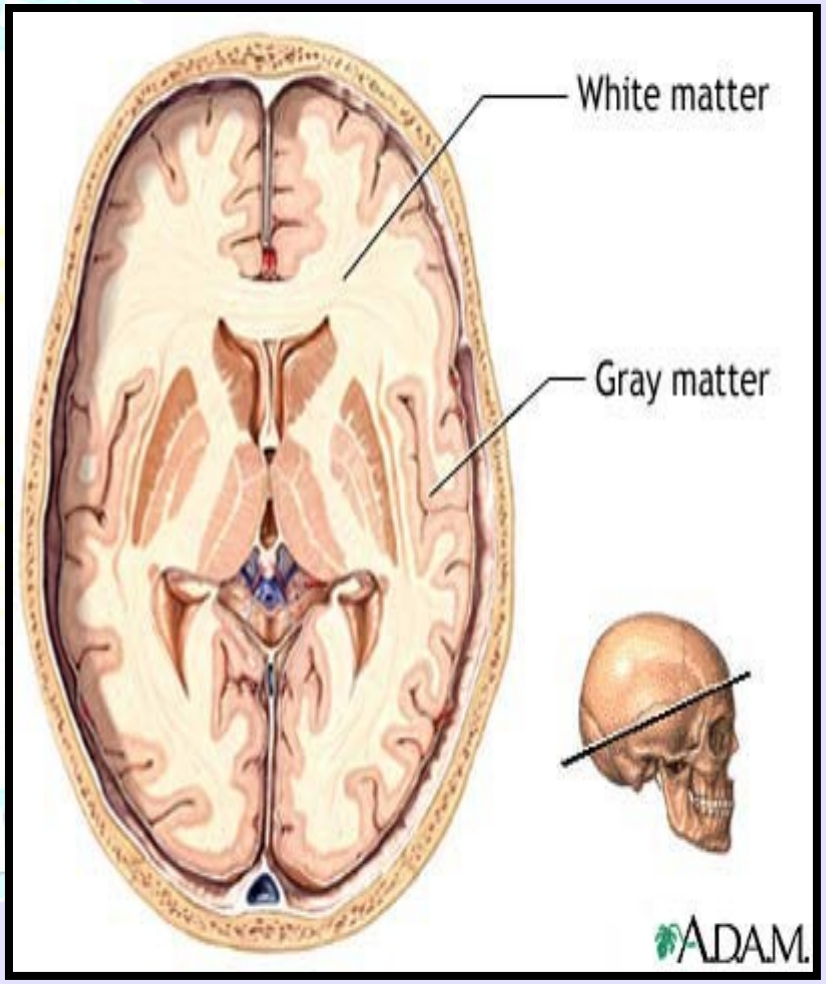
The sub thalamus lies between the mid brain and the thalamus, medial to the internal capsule and Globus Pallidus. It consist of:

- Grey Matter
- White Matter



GREY MATTER

- The cranial ends of the red nucleus and the substantia nigra extend into it.



WHITE MATTER

- Cranial ends of lemnisci, lateral to the red nucleus.
- Dentatothalamic tract along with the rubrothalamic fibers.
- Ansa Lenticularis (ventral)
- Fasciculus Lenticularis(dorsal)
- Subthalamic Fasciculus (intermediate fibers).

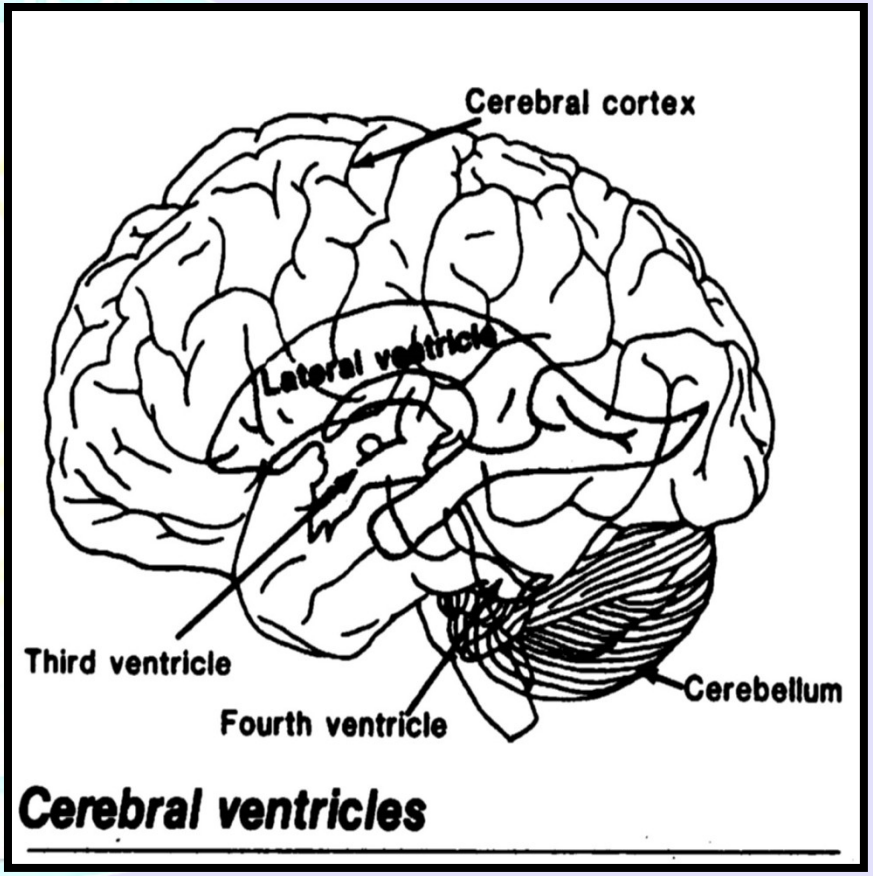


THIRD VENTRICLE

THIRD VENTRICLE

Location.

- The third ventricle is a narrow cavity located between the two hemispheres of the Diencephalon.



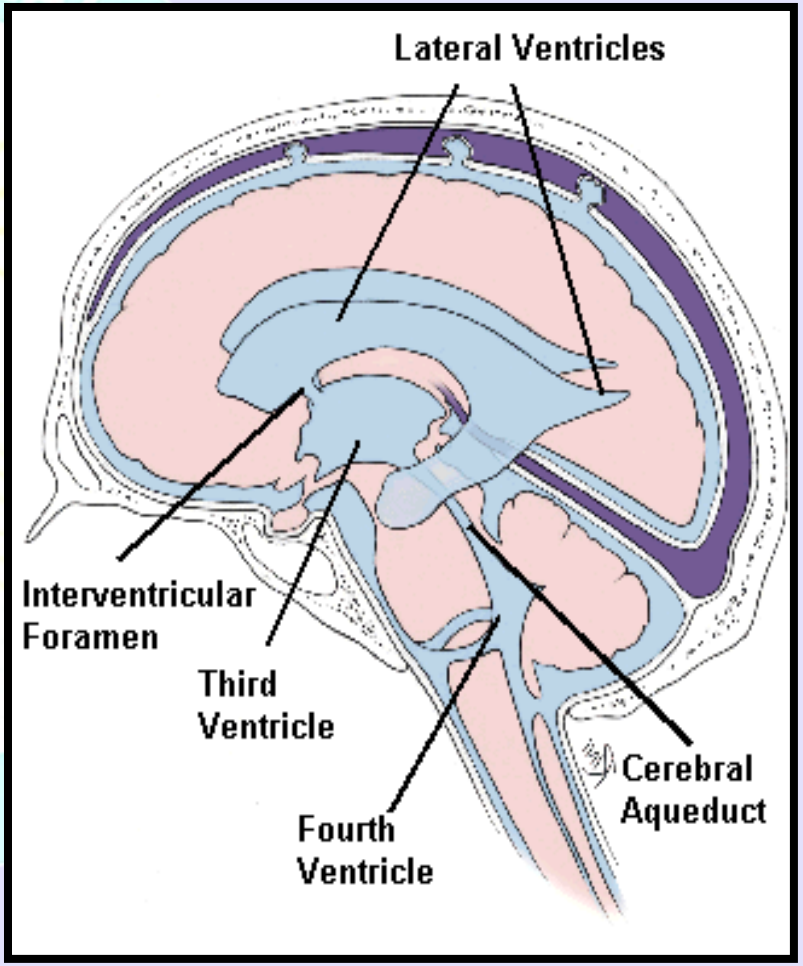
GROSS ANATOMY OF THIRD VENTRICLE

The third ventricle is the narrow vertical cavity of the diencephalon. A thin tela choroidea is formed in the roof of the third ventricle. The fornix and the corpus callosum are located superiorly. The lateral walls are formed by the medial thalamus and hypothalamus. The anterior commissure, the lamina terminalis, and the optic chiasm delineate the anterior wall. The floor of the third ventricle is formed by the infundibulum, which attaches the hypophysis, the tuber cinereum, the mammillary bodies, and the upper end of the midbrain. The posterior wall is formed by the pineal gland and habenular commissure.

THIRD VENTRICLE

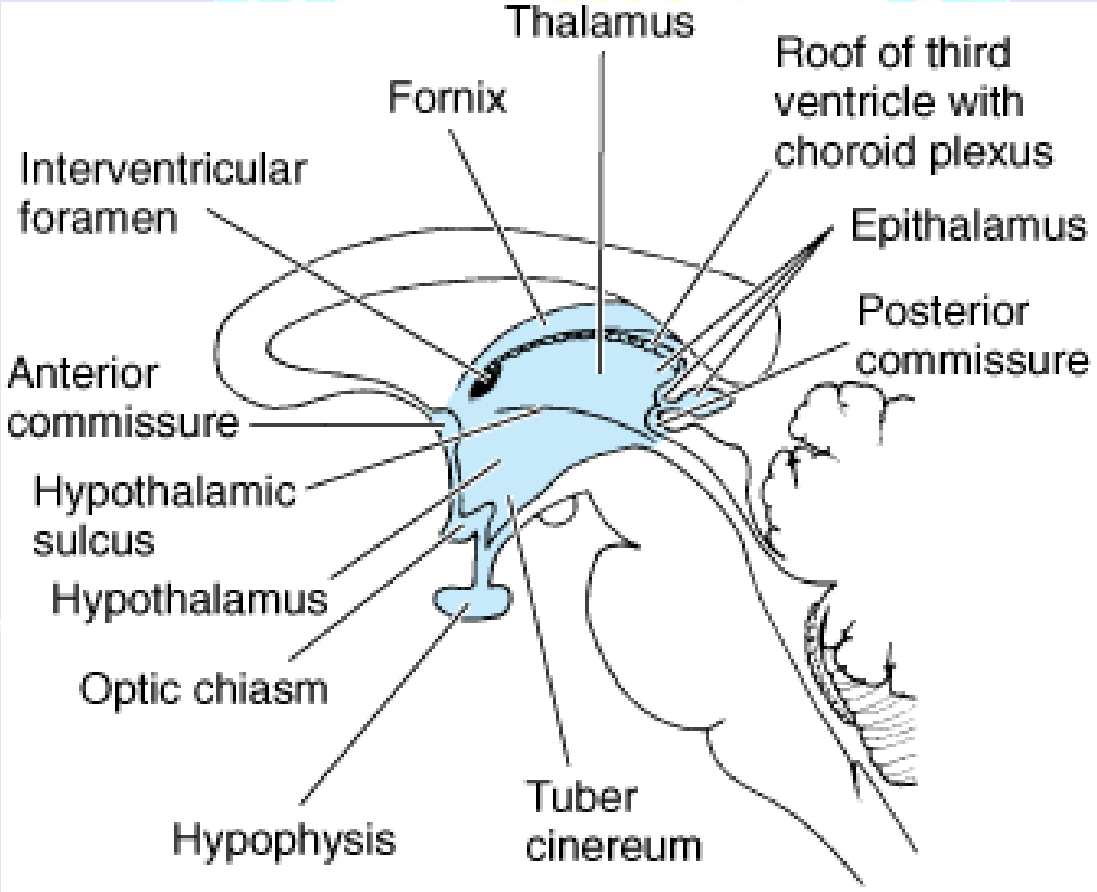
Function.

- Protects the brain from trauma
- Provides pathway for the circulation of cerebrospinal fluid



SELF ASSESSMENT

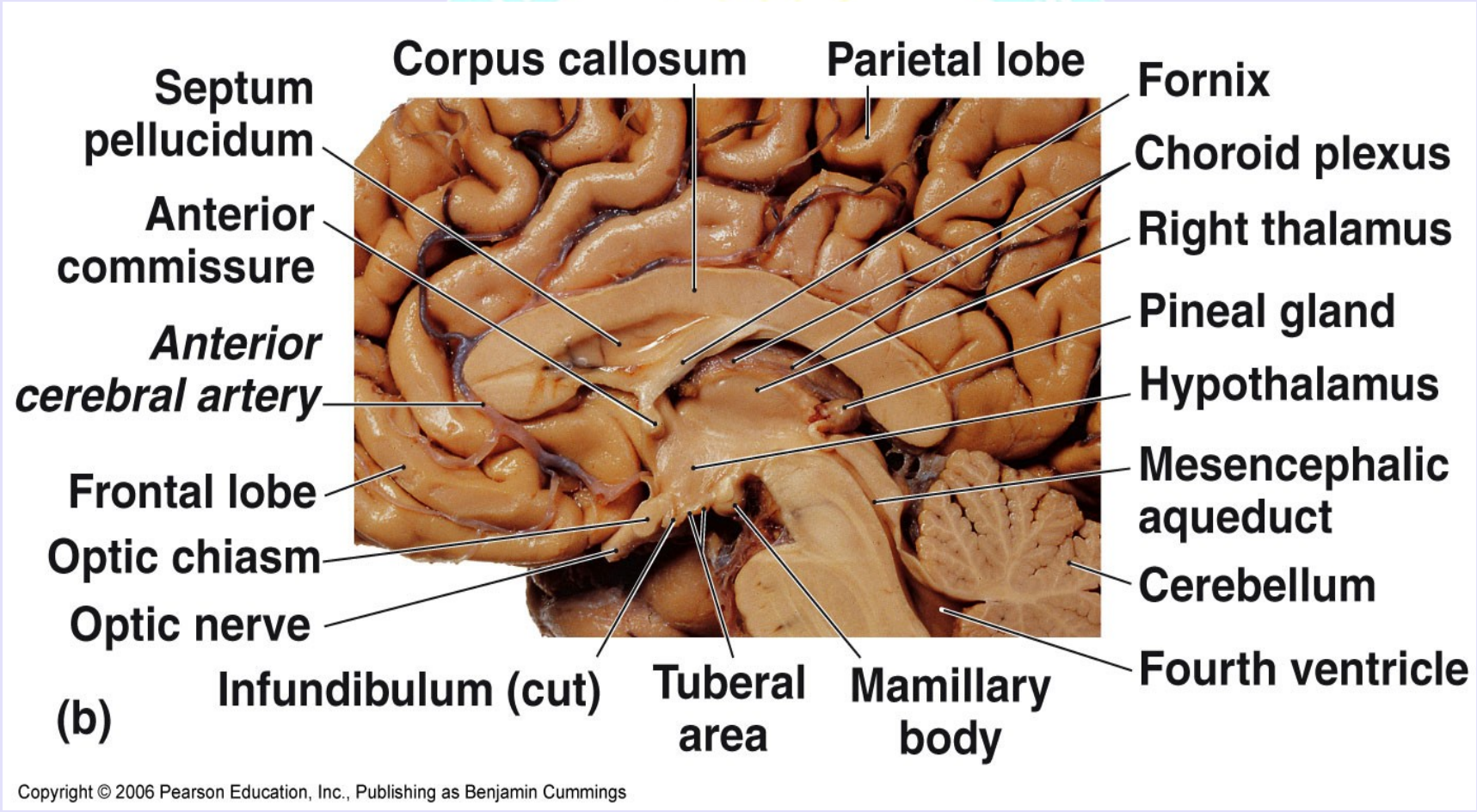
Q1.Identify the structure coloured in blue.



SELF ASSESSMENT

Q2.Which structures from the diencephalon are visible in this fig.

name them.



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SELF ASSESSMENT

- Q3.Name the groups of thalamic nuclei.
- Q4.Give the function of thalamus.
- Q5.What is the location of epithalamus in diencephalon?
- Q6.Which gland is the part of epithalamus? Give its functions.

THANK YOU